

Why Measure Theory

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1 Introduction: Who Cares

Write in the Introduction about why measure in general

2 Pointwise Convergence: Breaking at the Limit

2.1 Motivation for DCT

2.1.1 Pointwise Convergence

Consider the sequence of functions defined on $[0, 1]$

$$f_n(x) = \begin{cases} n : & 0 < x \leq \frac{1}{n} \\ 0 : & x = 0, \frac{1}{n} < x \leq 1 \end{cases} \quad (1)$$

First to show that $f_n(x) \rightarrow f(x) = 0$ pointwise consider the definition.

Definition 1 (Pointwise Convergence). For every $\epsilon > 0$ there exists some $N \in \mathbb{N}$ such that whenever $n \geq N$ it follows that

$$|f_n(x) - f(x)| < \epsilon$$

To prove the function converges pointwise I will show it satisfies the definition

Proof. Substituting our functions gives

$$|f_n(x) - 0| = 0 < \epsilon \text{ whenever } \frac{1}{n} < x$$

By the archimedean property we know for any $x \in \mathbb{R}$ there exists an $N \in \mathbb{N}$ such that whenever

$$\begin{aligned} n \geq N &\implies \frac{1}{n} < x \\ &\implies |f_n(x) - 0| = 0 \\ &\implies |f_n(x) - 0| < \epsilon \end{aligned}$$

For the special case of $x = 0$ observe that $f_n(0) = 0$ giving

$$\begin{aligned} f_n(0) = 0 &\implies |f_n(0) - 0| = 0 \\ &\implies |f_n(0) - 0| < \epsilon \end{aligned}$$

Meaning our function converges pointwise to $f(x)$ for $f(x) = 0$

□

Now to show the function does not converge uniformly first consider the definition of uniform convergence

Definition 2 (Uniform Convergence). For every $\epsilon > 0$ there exists some

$N \in \mathbb{N}$ such that whenever $n \geq N$ it follows that for all x

$$|f_n(x) - f(x)| < \epsilon$$

To show that $f_n(x) \not\rightarrow f(x)$ uniformly, I will show there exists no such N satisfying the definition for all x at once.

Proof. For contradiction assume there exists $N \in \mathbb{N}$ such that whenever $n \geq N$

$$|f_n(x) - f(x)| < \epsilon$$

for all x

Observe that for any such N

$$f_n(x) = n, \text{ for } 0 < x \leq \frac{1}{n}$$

Choose any $\epsilon \leq 1$ specifically I choose $\frac{1}{2}$ and call it ϵ_0 , then it is that case that for $0 < x \leq \frac{1}{n}$

$$|f_n(x) - 0| = n \implies |f_n(x) - 0| \geq \epsilon_0$$

But that's a contradiction since $f_n(x) \rightarrow f(x)$ uniformly gives an N that works for all x at once. Meaning $f_n(x) \not\rightarrow f(x)$ uniformly.

□

2.1.2 Limit Breaking

To show that

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx \neq \int_0^1 f(x) dx \quad (2)$$

Consider both methods of integration and there definitions

Definition 3 (Reimann Integration). Let f be a bounded function $f : [a, b] \rightarrow \mathbb{R}$. Let P be a partition of the domain of the form

$$P = \{x_0, x_1, \dots, x_n\}$$

Such that

$$a = x_0 < x_1 < \dots < x_n = b$$

If for every $\epsilon > 0$ there exists a partion P_ϵ such that $U(f, P_\epsilon) - L(f, P_\epsilon) < \epsilon$ then $f(x)$ is Riemann Integrable with $\int f = U(f) = L(f)$.

Where

$$U(f, P_\epsilon) = \sum_{k=1}^n (x_k - x_{k-1}) \sup_{[x_{k-1}, x_k]} f$$

$$L(f, P_\epsilon) = \sum_{k=1}^n (x_k - x_{k-1}) \inf_{[x_{k-1}, x_k]} f$$

and

$$U(f) = \inf_P U(f, P)$$

$$L(f) = \sup_P L(f, P)$$

Definition 4 (Lebesgue Integration). Let (X, S, μ) be a measure space and let $f(x) \rightarrow \mathbb{R}$ then $\int f = \sup L(f, P)$.

Where P is an S -Partition. Where an S -Partition is of the form

$$P = E_1, E_2, \dots, E_n$$

Where each $E_k \in S$ and

$$\bigcup_k E_k = X$$

and

$$\mathbb{L}(f, P) = \sum_k \mu(E_k) \inf_{E_k} f$$

Now I will show in accordance with the definitions neither integration method will suffice swapping the integral and the limit

Proof. First I will show that for any fixed n $f_n(x)$ is Riemann Integrable.

Fix $n \in \mathbb{N}$, first observe $|f_n(x)| \leq n$ for all $x \in [0, 1]$. Now take the partition where

$$P_n = \{0, \epsilon/8n, \frac{1}{n} - \epsilon/16n, \frac{1}{n} + \epsilon/16n, 1\} \text{ for } n > 1 \text{ and } P_1 = \{0, 1\}$$

Then summing length times sup on each interval

$$\begin{aligned}
U(f_n, P_n) &= (\epsilon/8n - 0)(n) \\
&\quad + ((\frac{1}{n} - \epsilon/16n) - (\epsilon/16n))(n) \\
&\quad + ((\frac{1}{n} + \epsilon/16n) - (\frac{1}{n} - \epsilon/16n))(0) \\
&= (\epsilon/8n)(n) + (\frac{1}{n} - \epsilon/16n)(n) + (\epsilon/16n)(0) \\
&= (\epsilon/8) + (1 + \epsilon/8) \\
&= 1 + \epsilon/4
\end{aligned}$$

and summing lengths times inf on each interval

$$\begin{aligned}
L(f_n, P_n) &= (\epsilon/8n - 0)(0) \\
&\quad + ((\frac{1}{n} - \epsilon/16n) - (\epsilon/16n))(n) \\
&\quad + ((\frac{1}{n} + \epsilon/16n) - (\frac{1}{n} - \epsilon/16n))(n) \\
&= (\epsilon/8n)(0) + (\frac{1}{n} - \epsilon/8n)(n) + (\epsilon/8n)(0) \\
&= 1 - \epsilon/8
\end{aligned}$$

Giving for every $\epsilon > 0$

$$U(f_n, P_n) - L(f_n, P_n) = 3\epsilon/8 < \epsilon$$

This gives

$$1 - \epsilon/8 \leq L(f_n) = \int_0^1 f_n = U(f_n) \leq 1 + \epsilon/4$$

Since $L(f_n, P_n) \leq L(f_n)$ and $U(f_n) \leq U(f_n, P_n)$ for all partitions P . Since the above equation holds for all $\epsilon > 0$ we have

$$\int_0^1 f_n = 1$$

for all fixed n

giving

$$\lim_{n \rightarrow \infty} \int_0^1 f_n = 1$$

Now taking any partition P will give

$$\int_0^1 f = 0$$

Since sup and inf will be 0 on any interval as $f(x) = 0$ everywhere.

This gives

$$\lim_{n \rightarrow \infty} \int_0^1 f_n = 1 \neq 0 = \int_0^1 f$$

□

(No matter how we integrate this its not going to work both riemann and lebesgue measure break here, its a motivation for DCT)

2.2 How Lebesgue Measure Explains the failure; and then saves us through DCT

(Show that for a function on $[0, 1]$ enumerating the rationals converging to dirchlet the function at the limit isnt in the class of riemann integrable

functions we know this because lebesgue defines whats riemann integrable; then we say that under DCT which is satisfies integrating with respect to measure we can swap the integral and the limit and it holds even though its pointwise)